UMET9U-30-1 Statistics Portfolio — Chat Notes

Countries selected: India (emerging, 1991–2024, N=34) and Germany (developed, 1961–2024, N=64)

Companion Excel file: `UMET9U-30-1_analysis_India_Germany.xlsx` — contains raw data, lagged variables, formula-driven descriptive statistics, correlations, both regressions, and all charts ready to copy into Word.

Part 1: Assignment Summary

Core details at a glance

- **Module:** UMET9U-30-1 — Professional Skills for Banking and Finance
- **Component:** Portfolio — Statistics component
- **Weighting:** 33% of the total module mark
- **Submission deadline:** Before **14:00 on 12/05/2026** (UK local time, 24-hour clock)
- **Late submission:** Eligible for the 48-hour late submission window
- **Marking/feedback returned by:** 11/06/2026
- **Word limit:** 1,500 words (includes headings, citations, quotes, lists; **excludes** statistical tables and reference list)
- **Format required:** Microsoft Word (.docx), 12-point Times New Roman or Arial
- **Submission method:** Electronic submission via Blackboard only (no email/paper/disk)

The topic

The assignment is built around Mauro (2003): **can lagged (previous-year) stock returns predict next-year economic growth, and does this relationship differ between emerging and developed economies?**

Financial markets react quickly to new information while macroeconomic variables react more slowly, so last year's stock returns may act as a leading indicator for this year's output growth.

Two variables:

- **r — Real stock returns** (%): log-return on stock index minus CPI inflation.
- **Dy — GDP per capita growth** (%): log-return on constant GDP per capita.

What you need to produce

A statistics report (⩽1,500 words) with six components:

1. **Short introduction** to the relationship between stock returns and economic growth (cite Mauro 2003).
2. **Descriptive statistics** for both variables in both countries — compute *and* discuss.
3. **Distributional analysis** using histograms and box plots.
4. **Relationship analysis** between lagged stock returns and economic growth (correlation + scatter plot).
5. **Regression model:** Dy on lagged r. Interpret slope, t-test, R^2. Run a **second regression** adding a lagged dependent variable as a robustness check.
6. **Short conclusion** summarising findings and comparing emerging vs developed results.

How the work is marked

Criterion	Weight
Data analysis (descriptives, regression, Excel graphics)	40%
Correct interpretation of statistical results	40%
Writing and presentation	20%

Formatting and first-page checklist

- Student number
- Module name and number (Professional Skills for Banking and Finance / UMET9U-30-1)
- Word count
- Coursework title
- Body in 12-point Times New Roman or Arial
- UWE Harvard referencing

Part 2: Worked Example with the Actual Data

I selected **India** (emerging, 1991–2024) and **Germany** (developed, 1961–2024). The numbers below come directly from the Blackboard dataset and all appear, as live formulas, in the companion Excel file.

Step 1 — Data setup

For each country I built five columns: `Year`, `r`, `Dy`, `r(-1)`, `Dy(-1)`. The lagged columns use the simple formula `=B{row-1}` and `=C{row-1}`. India yields **33 usable observations** for regression (one dropped to the lag); Germany yields **63**.

Step 2 — Descriptive statistics

Computed with `=AVERAGE()`, `=MEDIAN()`, `=STDEV()`, `=MIN()`, `=MAX()`, `=SKEW()`, `=KURT()`, `=COUNT()` referencing the data sheets.

Table 1. Descriptive statistics (%)

Statistic	India r	India Dy	Germany r	Germany Dy
Mean	5.534	4.395	2.947	1.954
Median	6.823	5.284	6.165	1.881
Std. deviation	27.238	2.862	22.595	2.202
Minimum	−81.319	−6.925	−59.300	−5.451
Maximum	50.370	8.426	48.872	6.363
Skewness	−0.919	−2.066	−0.578	−0.718
Excess kurtosis	1.935	6.522	0.208	1.514
Observations	34	34	64	64

What the numbers say:

- India has **higher average GDP growth** (4.40% vs 1.95%) — consistent with emerging-market catch-up growth.
- India's real stock returns average **5.53%**, Germany's **2.95%**; India is more volatile (SD 27.2% vs 22.6%) — the classic emerging-market risk-return profile.
- All four series are **negatively skewed**; India Dy has very high excess kurtosis (**6.52**) — rare but very deep recessions (e.g. the 1991 balance-of-payments crisis, 2020 lockdown).
- The minimums are striking: India r hit −**81.3%** and Germany r hit −**59.3%** (both around the 2008 GFC).

Step 3 — Histograms and box plots

In Excel: `Insert → Chart → Histogram` and `Insert → Chart → Box & Whisker`. The companion file has polished versions on the `Charts` sheet.

What to write about them:

- India's Dy histogram is sharply peaked around 5–7% with a long **left tail** from the 1991 and 2020 shocks; the box plot flags these as outliers below the lower whisker.
- Germany's Dy is more symmetric around 2% but also has two left-tail outliers (2009, 2020).

- The stock-return histograms are approximately symmetric around their means but fat-tailed — consistent with the literature on financial returns.

Step 4 — Correlation and scatter

Using `=CORREL(r_lag_range, Dy_range)`:

Table 2. Correlation between r(−1) and Dy

	India	Germany
corr(r(−1) , Dy)	**−0.065**	**+0.371**
Observations	33	63

This is the headline finding. **For Germany there is a clear positive relationship; for India the correlation is essentially zero** (slightly negative but economically negligible). The scatter plot makes this visual: the German cloud tilts upward with a visible trendline slope; the Indian cloud is flat.

Step 5 — Regression 1: Dy = α + β · r(−1)

Run with `Data Analysis → Regression`, or via `INTERCEPT`, `SLOPE`, and `LINEST` formulas.

Table 3. Regression 1

	India Coef	India SE	India t	India p	Germany Coef	Germany SE	Germany t	Germany p
Intercept	4.598	0.493	9.34	< 0.001	1.831	0.263	6.97	< 0.001
r(−1)	**−0.0064**	0.0177	**−0.36**	**0.720**	**+0.0361**	0.0116	**+3.12**	**0.003**
R^2	0.004				**0.138**			
N	33				63			

Interpretation in plain English:

- **Germany:** A 1 percentage-point increase in last year's real stock return is associated with **+0.036 pp** higher GDP growth this year. The t-statistic of 3.12 easily exceeds the 5% critical value (≈2.00), and p = 0.003 is well below 0.05 — we reject H_0: β = 0. Stock returns are a **statistically significant leading indicator of German GDP growth.** R^2 = 0.138 means 13.8% of Germany's GDP growth variation is explained by lagged stock returns alone — modest but economically meaningful.

- **India:** The slope of −0.006 is effectively zero and statistically insignificant ($t = -0.36$, $p = 0.72$). R^2 is only 0.4%. **Lagged stock returns have no predictive power for Indian GDP growth** in this sample.

Step 6 — Regression 2: $Dy = \alpha + \beta_1 \cdot r(-1) + \beta_2 \cdot Dy(-1)$ (robustness)

Add a column for lagged Dy; select both X-columns together when running the regression.

Table 4. Regression 2

	India Coef	India SE	India t	India p	Germany Coef	Germany SE	Germany t	Germany p
Intercept	4.332	0.894	4.84	< 0.001	1.214	0.350	3.47	0.001
r(−1)	**−0.0076**	0.0183	−0.41	0.681	**+0.0426**	0.0114	**+3.74**	**< 0.001**
Dy(−1)	0.0625	0.174	0.36	0.722	**+0.2998**	0.118	**+2.54**	**0.014**
R^2	0.008				**0.221**			
N	33				63			

Why this matters for the report:

- For **Germany**, adding lagged GDP growth controls for output persistence. Crucially, the coefficient on r(−1) *increases slightly* from 0.036 to **0.043** and remains highly significant ($t = 3.74$, $p < 0.001$). Lagged GDP growth itself is significant ($t = 2.54$), and R^2 rises from 0.138 to **0.221**. This robustness check **strengthens** the case that stock returns carry genuine leading information about the real economy, over and above the autoregressive component of GDP.
- For **India**, both lagged variables are insignificant and R^2 stays at just 0.8%. The null result is robust — stock returns simply don't lead Indian GDP growth at the annual frequency in this sample.

The headline finding

Mauro's (2003) prediction that stock returns are a less reliable leading indicator in emerging markets is strongly supported by this data. Germany shows a clear, robust, statistically significant relationship between last year's stock returns and this year's GDP growth. India — despite far more rapid growth — shows no such relationship. Possible reasons (to raise in the discussion): larger informal/agricultural sector, thinner market capitalisation relative to GDP, higher political and policy noise, and foreign-portfolio-flow-driven index movements that may be decoupled from domestic output.

Part 3: How to Turn This Into the Word Report

Here's a concrete plan for writing the 1,500-word report using the Excel file as your data layer.

Section-by-section plan with word budget

1. First page (not counted in word limit)

- Student number, module name & code, word count, coursework title. One clean line per item.

2. Introduction (~200 words)

Set up the leading-indicator hypothesis: stock markets price in news quickly, macro variables react with a lag, so last year's stock performance may forecast this year's GDP. Cite **Mauro (2003)** as the core reference and one or two others (Fama 1981 on stock returns and real activity; Stock & Watson on leading indicators). State that you compare one emerging and one developed economy (India and Germany) to test whether the relationship differs — a central question in Mauro. End with a one-sentence roadmap.

3. Data (~120 words)

Define r and Dy precisely (the variable table from the brief works). State sources (Bloomberg for indices, FRED for inflation and GDP per capita). Report the sample window for each country (India 1991–2024, N = 34; Germany 1961–2024, N = 64) and note the lag reduces each by one observation.

4. Descriptive statistics (~230 words)

Paste **Table 1** from the Descriptives sheet. Do not just list the numbers in prose — draw out three or four comparisons that matter:

- India's higher mean Dy vs Germany's — catch-up growth.
- Both stock-return series are far more volatile than their GDP-growth series (SDs of 27% and 23% vs 2.9% and 2.2%).
- India's Dy skewness of −2.07 and excess kurtosis of 6.52 flag extreme downside events (1991 crisis, 2020).
- Germany's return distribution is closer to symmetric (skew −0.58, kurtosis 0.21).

5. Distributional analysis (~200 words)

Insert **Figure 1 (histograms)** and **Figure 2 (box plots)** from the Charts sheet. Write two short paragraphs — one per variable type. For the histograms: comment on peak location, spread, and tails. For the box plots: comment on the IQR (the "box"), the whiskers, and any flagged outliers.

Connect the visual picture back to the skewness/kurtosis numbers in Table 1 — don't treat them as separate analyses.

6. Correlation and scatter (~200 words)

Paste **Table 2** and **Figure 3 (scatter plot)**. This is where the emerging-vs-developed contrast starts to show. Report both correlations (India −0.065, Germany +0.371). Describe what the scatters reveal: Germany's upward-sloping cloud vs India's flat cloud. Preview the regression section by noting that these correlations foreshadow the regression results.

7. Regression analysis (~420 words — the biggest section)

This is where 40% of the marks live. Paste **Table 3** (Regression 1) and **Table 4** (Regression 2).

For **Regression 1**, write one paragraph per country covering:

- The estimated slope and what a 1 pp change in r(−1) implies for Dy.
- The t-statistic and p-value, and the explicit conclusion about H_0: $\beta = 0$ at the 5% level.
- R^2 and what fraction of GDP-growth variation lagged returns explain.

For **Regression 2**, explain *why* you're adding Dy(−1) — output growth has inertia, so you want to check whether r(−1) still predicts Dy once you control for that persistence. Then report what happens:

- Germany: the r(−1) coefficient becomes *larger* and more significant, R^2 rises — robust.
- India: nothing changes; the null result is robust the other way.

Close this section with the comparative statement: the relationship hypothesised by Mauro (2003) holds clearly in Germany but not in India.

8. Conclusion (~150 words)

Summarise the headline finding (stock returns lead GDP growth in Germany; not in India). Offer one or two candidate explanations for the asymmetry (market depth, informal sector share, policy/political volatility). Then — this part is often forgotten and explicitly on the marking rubric — note **limitations and possible extensions**:

- Small samples, especially India (N = 33 for the regression).
- Results depend on the two countries chosen; a panel of all 20 countries would give more power.
- Possible structural breaks (1997 Asian crisis, 2008 GFC, 2020 pandemic) not addressed.
- No controls for monetary policy, commodity cycles, or global factors.
- Annual frequency may miss quarterly lead-lag dynamics.

9. Reference list (not counted in word limit)

Minimum four references in **UWE Harvard** format. Core ones:

- Mauro, P. (2003) 'Stock returns and output growth in emerging and advanced economies', *Journal of Development Economics*, 71(1), pp. 129–153.
- Fama, E.F. (1981) 'Stock returns, real activity, inflation, and money', *American Economic Review*, 71(4), pp. 545–565.
- Plus your data sources: Bloomberg, FRED (Federal Reserve Bank of St. Louis), with access dates.

Practical Word-document tips

- **Paste charts as pictures** (not linked objects) — prevents them breaking if the Excel file moves. Right-click chart in Excel → Copy → in Word, Paste Special → Picture (Enhanced Metafile or PNG).
- **Number your figures and tables** (Figure 1, Figure 2, Table 1, Table 2…) and caption each one. Reference them by number in the text: *"Figure 3 shows…"*.
- **Do not paste raw Excel output blocks.** Retype the important numbers into clean tables (Word's built-in table styles are fine). Markers can read cleaner tables faster and it looks more professional.
- **Numbers in the text should be consistent** with your tables. Round to 3 decimal places for coefficients and 2 decimal places for t-statistics and percentages.
- **Word count check:** highlight your body text only (excluding the first page, tables, and reference list) → bottom-left of Word shows the word count. Aim for **1,400–1,490** so you have headroom.

Pre-submission checklist

- [ ] First page with student number, module code, word count, title
- [ ] 12-point Times New Roman or Arial body text
- [ ] All tables numbered with captions
- [ ] All figures numbered with captions
- [ ] UWE Harvard references in alphabetical order
- [ ] Word count ≤ 1,500 (excluding tables, figures, references)
- [ ] Weaknesses/improvements paragraph in conclusion
- [ ] Saved as .docx
- [ ] Submitted via Blackboard before 14:00 on 12 May 2026
- [ ] Receipt kept